

AYAAN KUMAR

+1-(425)-588-1855 | ayaank1532@gmail.com | ayaank4@uw.edu | [Personal Website](#) | [GitHub](#) | [LinkedIn](#)

EDUCATION

University Of Washington, Seattle

Major: Bachelor of Science, Informatics

Sep 2024 – June 2028

- GPA: 4.0 | Dean's List 2024-2025
- Relevant Coursework: Calculus I, II, III, Intro to Computer Science, Data Structures & Algorithms I, II, Software Engineering, Discrete Math, Intro to AI, Computers & Society
- Clubs & Experience: AI Collective, DubHacks, Association for Computing Machinery, and HackRover.

SKILLS AND FRAMEWORKS

Programming Languages: Java, JavaScript, HTML, CSS, Python, C, C++, SQL, TypeScript

Frameworks: React, Node.js, TailwindCSS, Pandas, Flask

Tools & Technologies: AWS, Git, JSON, Linux, GitHub, MySQL, Firebase/Firestore, Docker

Concepts: Software development, Backend development, Object-Oriented Programming (OOP), Data Structures, Algorithms, Data Programming, Agile Methodologies, Machine Learning, Scalable Systems, RAG, Agentic AI, Deep Learning, Product Analytics

EXPERIENCE

Sensors, Energy, and Automation Laboratory (SEAL) – University of Washington

Seattle, WA

Associate Researcher & Developer

January 2026 – Present

- Collaborated with graduate researchers on a paper investigating human-computer interaction and data visualization methods for sustainable systems, contributing to methodology design and technical analysis.
- Implemented new interactive features for the SEAL Lab Dashboard, enhancing data accessibility and usability **for 150+ active users** through improved UI design and dynamic data visualization.
- Built scalable data pipeline and dashboard components that automated key data aggregation and visualization tasks, **improving research analysis efficiency by ~35%** and reducing repetitive manual reporting.

House of Wisdom

Redmond, WA

Software Engineering Intern

June - Sep 2025

- Spearheaded backend development for a tutoring management portal serving **200+ students and tutors**, enabling secure student-tutor communication, class registration, scheduling logistics, and attendance tracking with near **100% uptime**.
- Developed and deployed a **Firebase-based clock-in/clock-out system**, streamlining tutor shift tracking/payroll management and improving administrative efficiency by **35%**, saving over **10 staff hours per week**.
- Designed scalable **REST APIs** using **Flask** and **Firestore** to manage real-time class data, ensuring consistent and reliable backend performance while leveraging **Google Calendar** and **Google Sheets APIs** to automate scheduling, reducing work by **50%**.
- Built internal admin dashboards that visualized tutor hours, student attendance, and class trends, empowering staff to make data-driven adjustments that improved class fill rates by **20%**.

BAPA Consulting

Remote

Freelance Developer

Jun 2024 - Present

- Delivered over **15** responsive websites using HTML5, CSS3, JavaScript, and React, improving client website performance by **30%**, through lazy loading and efficient state management, and core web vitals across all projects.
- Increased average client user engagement by **40%** through intuitive, conversion-focused UI/UX designs aligned with each client's audience and content strategy while adhering to proper visual hierarchy and web design principles.
- Integrated **AWS Lambda** and **EmailJS** for automated contact forms and email notifications, boosting form submission conversions by **25%** and cutting client response time by **2 hours per query**.
- Implemented **TailwindCSS** for responsive, mobile-first interfaces, reducing front-end development time by **20%** and improving visual consistency across breakpoints.

PROJECTS

Career Quest – Interactive Career Planner for Kids (Typescript, Gemini API, React, Firebase, JSON)

- Built an AI-powered web app that helps children explore careers by tailored career simulations for **50+ unique professions**.
- Integrated **Gemini API** to produce recommendations and generate images boosting session engagement by **65%** during prototype with **10+ users**. Implemented efficient **JSON-based API handling** and **TypeScript frontend**, reducing load times by **35%**

VeriText (Python, Flask, NLP, Anthropic API, TypeScript)

- Built an **AI text detection service** using Python and Flask that performs **token-level NLP analysis** to highlight specific flagged phrases in submitted text, surfacing the exact source passages that triggered each detection rather than a binary verdict.
- Engineered a **React frontend with inline annotations rendering** that displays source citations alongside each flagged segment, allowing users to paraphrase flagged content directly; reduced manual review time by **~60%** compared to existing detection tools.

